

Abstract Algebra - Homework 2

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Problem 1

Write $|G| = p^\alpha m$ with $p \nmid m$ and $|P| = p^\alpha$. Because $N \trianglelefteq G$, Lagrange gives $|N| = p^\beta d$ where $0 \leq \beta \leq \alpha$ and $p \nmid d$. First consider $N \cap P$. Since $N \cap P \leq P$, we have

$$|N \cap P| \mid |P| = p^\alpha \implies |N \cap P| = p^\gamma$$

for some $0 \leq \gamma \leq \beta$; thus $N \cap P$ is a p -subgroup of N .

Next form $NP \leq G$. Using $|NP| = \frac{|N||P|}{|N \cap P|}$ we compute

$$|NP| = \frac{p^\beta d p^\alpha}{p^\gamma} = p^{\alpha+\beta-\gamma} d, \quad [NP : P] = \frac{|NP|}{|P|} = p^{\beta-\gamma} d.$$

Because $P \leq NP \leq G$, the index $[NP : P]$ divides $[G : P] = m$; since $p \nmid m$ we must have $p^{\beta-\gamma} = 1$. Hence $\beta = \gamma$ and $|N \cap P| = p^\beta$. This shows $N \cap P$ attains the maximal power of p dividing $|N|$, so $N \cap P$ is a Sylow p -subgroup of N .

For the quotient G/N we have

$$|G/N| = \frac{|G|}{|N|} = \frac{p^\alpha m}{p^\beta d} = p^{\alpha-\beta} \frac{m}{d},$$

so any Sylow p -subgroup of G/N has order $p^{\alpha-\beta}$. Since $N \leq NP$, the subgroup $NP/N \leq G/N$ satisfies

$$|NP/N| = \frac{|NP|}{|N|} = \frac{p^\alpha d}{p^\beta d} = p^{\alpha-\beta},$$

hence NP/N is a Sylow p -subgroup of G/N .

Thus $N \cap P$ is Sylow- p in N , and NP/N is Sylow- p in G/N .

Problem 2

- (1) Let $N = \ker f$. For any subgroup $S \leq G$ one has

$$f^{-1}(f(S)) = NS,$$

because every element mapping into $f(S)$ differs from an element of S by an element of N .

Assume $f(P) = f(Q)$. Then

$$P, Q \subseteq f^{-1}(f(P)) = NP,$$

so P and Q are both Sylow p -subgroups of the same subgroup NP . Sylow's conjugacy theorem applied inside NP gives an element $x \in NP \subseteq G$ with $xPx^{-1} = Q$. Hence P and Q are conjugate in G .

- (2) For the inclusion $f(N_G(P)) \subseteq N_H(f(P))$ let $g \in N_G(P)$; then $gPg^{-1} = P$ and

$$f(g)f(P)f(g)^{-1} = f(gPg^{-1}) = f(P),$$

so $f(g) \in N_H(f(P))$.

Conversely, let $h \in N_H(f(P))$. Choose $g \in G$ with $f(g) = h$. Since $f(gPg^{-1}) = f(g)f(P)f(g)^{-1} = f(P)$, part (1) implies gPg^{-1} is conjugate to P in G : there exists $x \in G$ with $xgPg^{-1}x^{-1} = P$. Put $y = xg$; then $y \in N_G(P)$ and $h = f(g) = f(x)^{-1}f(y) \in f(N_G(P))$. Hence $N_H(f(P)) \subseteq f(N_G(P))$.

Therefore $f(N_G(P)) = N_H(f(P))$.

- (3) If $p \nmid |H|$ then $|f(P)| \mid |H|$ and $|f(P)|$ is a power of p ; hence $|f(P)| = 1$ and $f(P) = \{e\} \subseteq \ker f$. From part (2) we have $f(N_G(P)) = N_H(f(P)) = H$, because the normaliser of the trivial subgroup is all of H .

Problem 3

- (1) (a) By Sylow's theorems, the number n_5 of 5-Sylow subgroups satisfies $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid \frac{|G|}{p^k} = \frac{15}{5} = 3$, so $n_5 = 1$. A unique Sylow subgroup is always normal, so G has a normal subgroup of order 5.

(b) Let P_3 and P_5 be the Sylow 3- and 5-subgroups of G , with $|P_3| = 3$ and $|P_5| = 5$. Since $|P_3| = 3$ and $|P_5| = 5$ are relatively prime, and $|P_3||P_5| = 15 = |G|$, it follows that $P_3 \cap P_5 = \{e\}$. Moreover, P_5 is normal in G by part (a), and both P_3 and P_5 are cyclic because all groups of prime order are cyclic. Therefore, the internal direct product P_3P_5 is a subgroup of G of order 15. Since its order equals $|G|$, we have:

$$G = P_3P_5 \cong P_3 \times P_5 \cong \mathbb{Z}/3 \times \mathbb{Z}/5 \cong \mathbb{Z}/15,$$

so G is cyclic.

- (2) Again by Sylow's theorems, the number n_p of p -Sylow subgroups satisfies $n_p \equiv 1 \pmod{p}$ and divides q , so $n_p = 1$ or q . Similarly, $n_q \equiv 1 \pmod{q}$ and divides p , so $n_q = 1$ or p . Suppose for contradiction that $n_q = p$. Then:

$$p \equiv 1 \pmod{q} \Rightarrow p \equiv 1 \pmod{q},$$

which contradicts the assumption $p \not\equiv 1 \pmod{q}$. Therefore, $n_q = 1$, and the q -Sylow subgroup is unique and hence normal.

All groups of prime order are cyclic, so both the p -Sylow and q -Sylow subgroups are cyclic. Let P and Q denote the p - and q -Sylow subgroups, with $Q \trianglelefteq G$ and $P \cap Q = \{e\}$. Since $|P||Q| = pq = |G|$ and Q is normal, we have:

$$G \cong P \times Q \cong \mathbb{Z}/p \times \mathbb{Z}/q \cong \mathbb{Z}/pq,$$

because p and q are relatively prime. Hence, G is cyclic.

- (3) (a) Once again by Sylow's theorems, the number n_3 of 3-Sylow subgroups satisfies $n_3 \equiv 1 \pmod{3}$ and divides 4. Thus, $n_3 = 1$ or 4. The number n_2 of 2-Sylow subgroups satisfies $n_2 \equiv 1 \pmod{2}$ and divides 3. So $n_2 = 1$ or 3. If either $n_3 = 1$ or $n_2 = 1$, the corresponding Sylow subgroup is unique and hence normal. If both were non-unique (i.e., $n_3 = 4$ and $n_2 = 3$), then G would contain more than 12 elements, which is impossible. Hence, at least one of the Sylow subgroups is normal and therefore, G always has a normal subgroup.

(b) A non-cyclic group of order 12 is the alternating group A_4 , the group of even permutations on 4 elements. It has order 12 but is not

cyclic. A cyclic group of order 12 must have an element of order 12. However, in A_4 , all elements have order 1, 2, or 3. Therefore, A_4 cannot be cyclic.

Problem 4

First, note that both a and b are products of two disjoint 4-cycles. Therefore,

$$|a| = |b| = \text{lcm}(4, 4) = 4.$$

Since their cycles overlap, they do not act on disjoint sets, so we compute explicitly:

$$ab = (1234)(5678)(1537)(2648), \quad ba = (1537)(2648)(1234)(5678).$$

We compute

$$ab = (17)(28)(36)(45) = ba,$$

so a and b commute and H is abelian.

To determine the order of H , observe that a has order 4, and b also has order 4. We compute a^2 , b^2 , and their combinations to find the number of distinct elements generated. It turns out that:

$$H = \langle a, b \rangle = \{e, a, a^2, a^3, b, ab, a^2b, a^3b\}$$

which contains 8 distinct elements. So $|H| = 8$.

Since H is a finite abelian group of order 8, by the classification theorem for finite abelian groups, it must be isomorphic to one of:

$$\mathbb{Z}_8, \quad \mathbb{Z}_4 \times \mathbb{Z}_2, \quad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

We note that H contains elements of order 4 (e.g., a), but no element of order 8, so it cannot be cyclic. It also contains an element of order 4 and another of order 2 (e.g., b^2), so it is not the elementary abelian group $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, which has all elements of order 2.

Therefore, the group must be:

$$H \cong \mathbb{Z}_4 \times \mathbb{Z}_2.$$

Problem 5

- (1) Assume G/Z is cyclic, so there exists $g \in G$ such that every element of G/Z is of the form $g^n Z$ for some $n \in \mathbb{Z}$. Then for any $x \in G$, there exists $n \in \mathbb{Z}$ and $z \in Z$ such that:

$$x = g^n z.$$

Now let $x = g^n z$ and $y = g^m z'$, with $z, z' \in Z$. Then:

$$xy = g^n z \cdot g^m z' = g^{n+m} z z', \quad \text{and} \quad yx = g^m z' \cdot g^n z = g^{m+n} z' z.$$

Since $Z \subseteq Z(G)$, we have $z z' = z' z$, so:

$$xy = yx.$$

Thus, any two elements of G commute, so G is abelian.

- (2) Since G is a p -group, its center $Z(G)$ is nontrivial. Hence, $|Z(G)| = p$ or p^2 . If $|Z(G)| = p^2$, then $Z(G) = G$, so G is abelian. If $|Z(G)| = p$, then the quotient group $G/Z(G)$ has order p and is cyclic. By previous section, if $G/Z(G)$ is cyclic and $Z(G)$ is central, then G is abelian. Therefore, in both cases, G is abelian.
- (3) Let p be a prime. Up to isomorphism, there are exactly two groups of order p^2 :

$$\mathbb{Z}_{p^2}, \quad \mathbb{Z}_p \times \mathbb{Z}_p.$$

These are the only abelian groups of order p^2 , and by previous section, all groups of this order are abelian. Therefore, this list is complete up to isomorphism.

Problem 6

- (1) Since $A \cong \mathbb{Z}_n$, any homomorphism $\chi \in \widehat{A}$ is determined by the image of a generator, and this image must be an n -th root of unity. Therefore, $\widehat{A} \cong \mu_n$, the group of n -th roots of unity, which is cyclic of order n . Hence, $\widehat{A}$ is cyclic of order n .

- (2) By the structure theorem for finite abelian groups, A is isomorphic to a product of cyclic groups:

$$A \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}.$$

Then,

$$\widehat{A} \cong \widehat{\mathbb{Z}_{n_1}} \times \cdots \times \widehat{\mathbb{Z}_{n_k}},$$

and each $\widehat{\mathbb{Z}_{n_i}} \cong \mathbb{Z}_{n_i}$. Therefore, $\widehat{A} \cong A$, so $|\widehat{A}| = |A| = n$.